Vijay Sundar Nagamalla

Forward Deployed Engineer (FDE) | AI Solutions and Technical Enablement
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PROFESSIONAL SUMMARY

AI application builder and technical educator targeting Forward Deployed Engineer roles. Brings 8+ years of working directly with clients and stakeholders at 40+ institutions, teaching programming (C, Java, Python, SQL, DSA) to 50,000+ learners, together with hands-on delivery of AI and full-stack applications (RAG, MCP tool calling, on-device speech-to-text) in Python, React, and FastAPI. Strong at turning complex technology into working software, live demos, and user training.

SKILLS

AI / LLM: RAG Pipelines, Embeddings and Vector Search (FAISS), MCP Tool Calling, LangChain, Claude API, NVIDIA NIM, Speech-to-Text (Whisper)

Client Enablement: Stakeholder Management, Technical Training, Workshops and Live Demos, Curriculum Design, Technical Writing, Public Speaking

Languages: Python, JavaScript, TypeScript, Java, C#, SQL

Frontend: React, TanStack Start, Tailwind CSS, shadcn/ui

Backend: FastAPI, Node.js / Express, .NET 8, REST APIs, JWT Auth, RBAC

Data and DevOps: PostgreSQL, MySQL, Redis, Supabase, Docker, Git, GitHub Actions, Vercel, Power BI

PROFESSIONAL EXPERIENCE

Independent Developer and Technical Trainer January 2025 – Present
Freelance

- Built six AI and full-stack applications (Almaanac, Vukti, PropIQ, ILFU, Forward, FDE Flow Atlas) with React, FastAPI, PostgreSQL, Docker, and the Claude and NVIDIA NIM APIs
- Run own training batches on Almaanac and Vukti to track learners' coding practice across platforms and to conduct MCQ, coding, and SQL assessments
- Authored a 133-topic interactive curriculum on AI application engineering (RAG, tool calling, agent workflows) with animated diagrams, quizzes, and 11 hands-on projects
- Deliver technical training in Java, Python, SQL, and DSA, and report learner outcomes through Power BI dashboards

Technical Project Manager May 2023 – December 2024
Xplore.co.in

- Owned large-scale upskilling programs as project owner and lead trainer, with 100% on-time delivery of technical training and labs
- Built and maintained Power BI dashboards that leadership used to monitor KPIs, learner engagement, and program health
- Ran program operations end to end: batch planning, stakeholder coordination, and multi-track logistics with instructional designers

Senior Technical Trainer May 2020 – April 2023
Freelance

- Trained 25,000+ learners in Java, Python, SQL, and DSA, with ~50% average improvement in assessment scores
- Managed learning operations for multi-track programs: session scheduling, batch planning, enrollment, and attendance
- Tracked completion rates and KPIs in MS Excel and Power BI, keeping operational data at 95%+ accuracy

Associate Trainer June 2018 – May 2020
FACE

- Taught Aptitude, C, Java, Python, and SQL to batches of 80+ learners
- Wrote lesson plans, presentations, and hands-on assessments, and gave structured feedback for interview readiness

KEY PROJECTS

PropIQ — AI Loan Intelligence Platform (RAG + MCP Tool Calling) GitHub
React, Node.js / Express, MySQL, FastAPI, LangChain, NVIDIA NIM (Llama 3.1), FAISS

- Full-stack platform simulating commercial real estate loan servicing, with a RAG pipeline over loan documents (FAISS vector store, NVIDIA embeddings, LangChain)
- AI copilot that routes each question to RAG or to one of four structured MCP tools (loan details, risk score, payment history, risky loans)
- Document intelligence (PDF upload with AI field extraction), rule-based risk scoring, and a per-loan default-probability and 6-month revenue forecast

Vukti — Academic Platform and Student OS vukti.in

React, Vite, Tailwind CSS, FastAPI, SQLAlchemy, PostgreSQL, Redis, Docker, Judge0, Monaco Editor

- Nine-module platform for technical institutions covering academics, placements, and practice; used with own training batches
- Examination engine for MCQ, coding, and SQL tests on a self-hosted, Docker-isolated Judge0 (25+ languages) with tab-switch proctoring
- Gamified coding practice with 1,000+ problems, a Monaco-based editor, streaks, levels, and leaderboards; syncs activity from six coding platforms through an async worker pool

Almaanac — Classroom Coding Analytics almaanac.vercel.app

TanStack Start (React), TanStack Query, Supabase (PostgreSQL, RLS), Tailwind CSS, Recharts, TypeScript, Vercel

- Aggregates learners' coding activity from eight platforms (LeetCode, Codeforces, CodeChef, HackerRank, and others) into one educator dashboard; used with own training batches
- Independent async refresh workers per platform with rate limiting, lease-based Postgres locking, and failure isolation
- Difficulty-weighted scoring, leaderboards, "At Risk" cohorts, 365-day heatmaps, and term-end PDF reports

FDE Flow Atlas — Interactive AI Engineering Curriculum myviapps.github.io/fde-flow-atlas

HTML, CSS, JavaScript, SVG Animation, GitHub Pages

- 133 topics on RAG pipelines, tool calling, and autonomous workflows, each taught through an animated flow diagram with step-by-step narration and code highlighting
- Includes case studies, quizzes, fill-in templates, and 11 hands-on projects, with learner progress saved in the browser

ILFU (I Listen For U) — Open-Source Local Meeting Transcriber GitHub

Python, faster-whisper, whisper.cpp, PyAudioWPatch (WASAPI), Pytest, AGPL-3.0

- Meeting transcriber that runs fully on the user's machine: no API keys, no accounts, and no bot joining the call (in active development)
- Captures microphone and speaker audio through WASAPI loopback, then runs on-device speech-to-text, speaker diarization, and local LLM summaries

Forward — Presenter and Live-Teaching Copilot (Windows)

C# WPF, .NET 8, Claude API, DeepSeek, NVIDIA NIM, NAudio, Whisper ONNX

- Desktop copilot that gives trainers and presenters live transcription and AI-generated speaker notes during a session, kept in a private presenter-only window
- Provider fallback chain (Claude, DeepSeek, NVIDIA NIM, Groq, Ollama) with automatic failover; Whisper speech-to-text runs on-device on an AMD NPU

EDUCATION

Bachelor of Technology (B.Tech) in Electronics and Communication Engineering Graduated 2019

North East Christian University